TERRAFORM DRIFT

WHAT IS TERRAFORM DRIFT

Terraform drift is a difference between the desired infrastructure defined in Terraform and the actual infrastructure running in the cloud.

1.main.tf

```
resource "aws_instance" "web" {
  instance_type = "t3.micro"
}
```

2.terraform apply

Terraform expects:
EC2 instance --> t3.micro

But someone manually changes the EC2 instance from AWS Console:
t3.micro → c7i-flex.large

1. Terraform expects → t3.micro
2. AWS actually has → c7i-flex.large

And this concept we called it as terraform drift

WHY DOES DRIFT HAPPENS

Drift commonly happens when:

- Someone manually changes AWS resources.
- Someone deletes a resource from the AWS Console.
- Another tool/automation changes the resource.
- Infrastructure is modified outside Terraform.

HOW TO DETECT DRIFT

1. Terraform plan

Terraform compares the configuration with the current infrastructure and shows the differences.

Terraform expects:
t3.micro

AWS:
C7i-flex.large

Difference detected!

HOW TO MANAGE DRIFT

Situation 1: We DON'T want the manual change

Terraform says: t3.micro
AWS has: c7i-flex.large

We want t3.micro

1) terraform plan
2) Terraform apply

Terraform brings the infrastructure back toward the configuration.

AWS has - c7i-flex. large ---> terraform apply ---> AWS - t3.micro

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Situation 2: We Want the manual change

AWS has ---> c7i-flex.large
We decide that c7i-flex.large is actually required.

1. Update main.tf

```
resource "aws_instance" "web" {
  instance_type = "c7i-flex.large"
}
```

2. terraform plan

Now Terraform configuration and AWS infrastructure agree.

1) Terraform → c7i-flex.large
2) AWS → c7i-flex.large

No Drift happens at these time

3. terraform apply
4. terraform state list